Dijkgraaf-Witten theories in 3d and boundaries

(w/ Fuchs, Valentino 1307.3632)

1. Some basic ideas
2. Boundary conditions
3. Outlook

1. Basic idea

1.1 M_3, G finite group

$$\mathrm{Eft}_G(M) = \int_{\mathrm{Bun}_G(M)} e^0 = |\mathrm{Bun}_G(M)|$$

finite groupoid

Def groupoid cardinality $|X|$

1. Homotopy invariance $X \cong Y \Rightarrow |X| = |Y|$
2. Gluing $|X \sqcup Y| = |X| + |Y|$
3. Normalization $|*/\!/\mathrm{id}\,|| = 1$

4. Covering $\quad F: X \to Y$ n-sheeted covering $\Rightarrow |X| = n\,|Y|$

Covering
- surjective on objects
- path lifting property $y_1 \xrightarrow{p} y_2$ s.t. $F(x_1) = y_1$

$\Rightarrow \exists!\ x_1 \xrightarrow{p'} x_2$ s.t. $F(p') = p$

Consequences:

- $$|\Gamma| = \sum_{[g] \in \pi_0(\Gamma)} \frac{1}{|\mathrm{Aut}(g)|}$$

- $$|X \times Y| = |X| \cdot |Y|$$

- $$|S /\!/ G| = \frac{|S|}{|G|} \quad \text{(orbit stabilizer thm)}$$

Thus $\quad \mathrm{Eff}_G(M) = |\mathrm{Bun}_G(M)| = \dfrac{|\mathrm{Hom}(\pi_1(M), G)|}{|G|}$

Trivially, invariant of manifolds. Main point: locality!

1.2 DW as (extended) TFT

<u>d=2</u> Σ_2. For M_3 s.t. $\partial M = \Sigma$ and fixed $P \in \mathrm{Bun}_G(\Sigma)$

$$\psi_M : \quad P \longrightarrow |\mathrm{Bun}_G(M, P)|$$

bundles restricting to P on Σ

gauge invariant function on $\mathrm{Bun}_G(\Sigma)$. Hence we assign to Σ:

$$\mathcal{H}_G(\Sigma) = \mathbb{C}\langle \pi_0(\mathrm{Bun}_G(\Sigma))\rangle \qquad \text{vector space}$$

Vector spaces = linearizations of categories of bundles for surfaces

<u>$d=1$</u> S_1 For Σ_2 s.t. $\partial\Sigma = S$ consider

$$\psi_{\Sigma} : \mathrm{Bun}_G(S) \to \mathrm{vect}$$

$$P \longmapsto \mathbb{C}\langle \pi_0(\mathrm{Bun}_G(\Sigma)), P\rangle$$

Vector bundle over $\mathrm{Bun}_G(S)$

$$\mathcal{H}_G(S) = [\mathrm{Bun}_G(S), \mathrm{vect}] \qquad \text{linear category}$$

In general

$$\mathrm{Cob}_{3,2,1} \xrightarrow{\widetilde{\mathrm{Bun}}} \mathrm{Span\,Grp} \xrightarrow{[-,\mathrm{vect}]} 2\text{-vect}$$

linearization, cocycles for proj. reps

1.3 Groupoid cohomology

$\omega \in Z^3(G, \mathbb{C}^*)$ gives more general DW categories

Groupoid cohomology $(\Gamma_0, \Gamma_1) \rightsquigarrow$ nerve

$$\Gamma_\bullet = \left(\cdots \overset{\rightarrow}{\underset{\rightarrow}{\rightrightarrows}} \Gamma_2 \overset{\rightarrow}{\underset{\rightarrow}{\rightrightarrows}} \Gamma_1 \rightrightarrows \Gamma_0 \right)$$

$\rightsquigarrow$ complex. Alternating combinations of face maps:

(generalizes bar complex)

$$\mathrm{Map}(\Gamma_0, \mathbb{C}^*) \to \mathrm{Map}(\Gamma_1, \mathbb{C}^*) \to \cdots$$

<u>Algebra</u>: $\tau \in Z^2(\Gamma, \mathbb{C}^*)$ gives a proj. repn. of Γ

$$\begin{tikzcd}
 & V_{h_2 g} \arrow[dr, "\rho_{h_1}"] \arrow[d, phantom, "\Downarrow \tau"] & \\
V_\gamma \arrow[ur, "\rho_{h_2}"] \arrow[rr, "\rho_{h_1 h_2}"'] & {} & V_{h_1 h_2 \gamma}
\end{tikzcd}$$

$$\rho_{h_1 \cdot h_2} = \tau(h_1, h_2, \gamma)\, \rho_{h_1} \circ \rho_{h_2}$$

<u>Geometry</u>:

<u>Def</u> Call $\omega \in Z^{n+1}(\Gamma, \mathbb{C}^*)$ an n-<u>gerbe</u> on the groupoid Γ.

n gerbes form an $(n+1)$-category

<u>Ex</u>

- 0-gerbe = line bundles. Objects $\omega \in Z^1(\Gamma)$, morphisms

$$\mathrm{Hom}(\omega_1, \omega_2) = \{ \eta \in \Gamma_0 \text{ s.t. } d\eta = \omega_1 \cdot \omega_2^{-1} \}$$

- 1-gerbe objects $\omega \in Z^2(\Gamma)$

  1-morph $\omega_1 \xrightarrow{\eta} \omega_2$ $\quad \eta \in \Gamma_1$ s.t. $d\eta = \omega_1 \omega_2^{-1}$

  2-morph $\eta_1 \Rightarrow \eta_2$

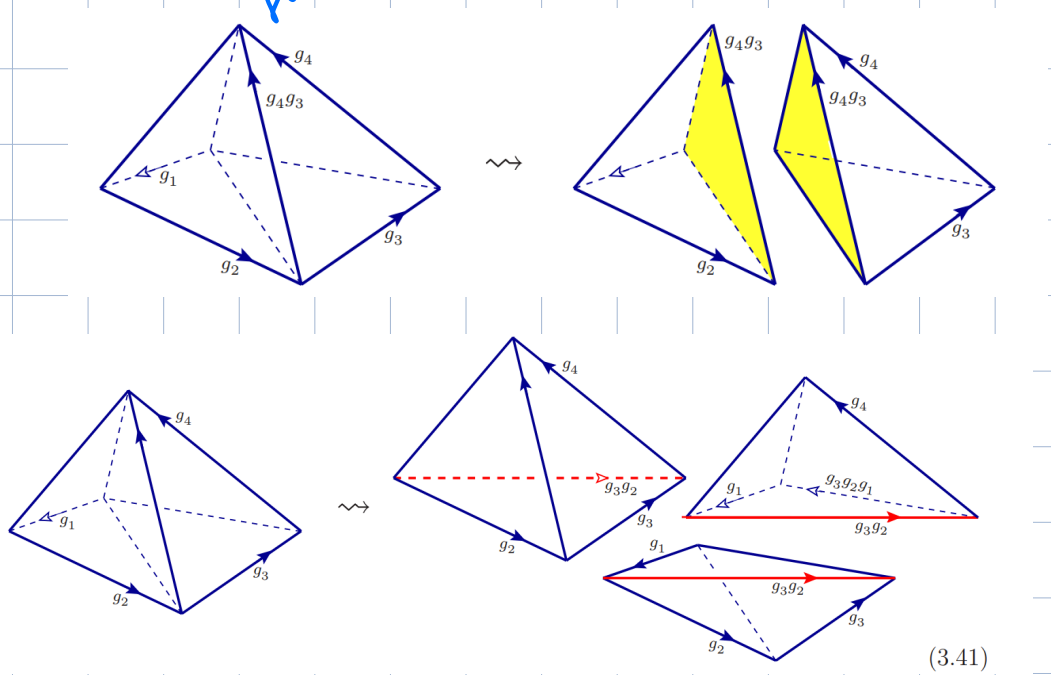
  $\phi \in \Gamma_0 \qquad d\phi = \eta_2 \eta_1^{-1}$

1.4. DW theory in the bulk

Chern Simons 2-gerbe $\omega \in Z^3(G, \mathbb{C}^*)$

$\rightsquigarrow$ topological action $S_\omega[M]$ "holonomy of 2-gerbe"

Well-defined:



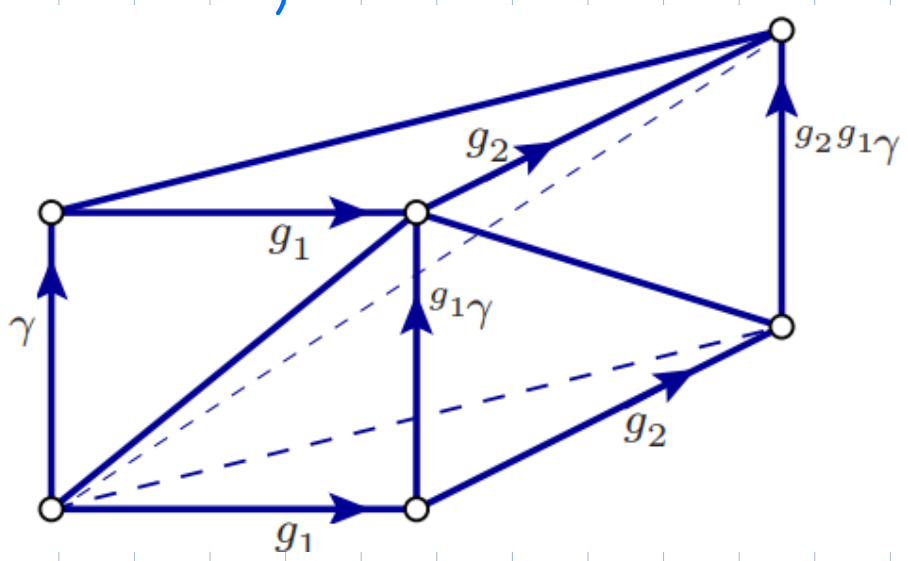
(3.41)

$\Leftrightarrow$ ω closed

$\Leftrightarrow$ pentagon

$\mathrm{Bun}_G(S^1) \cong G/\!/G$. Want projective reps. Transgress ω to

$$\tau(\omega) \in Z^2(G/\!/G, \mathbb{C}^\times)$$

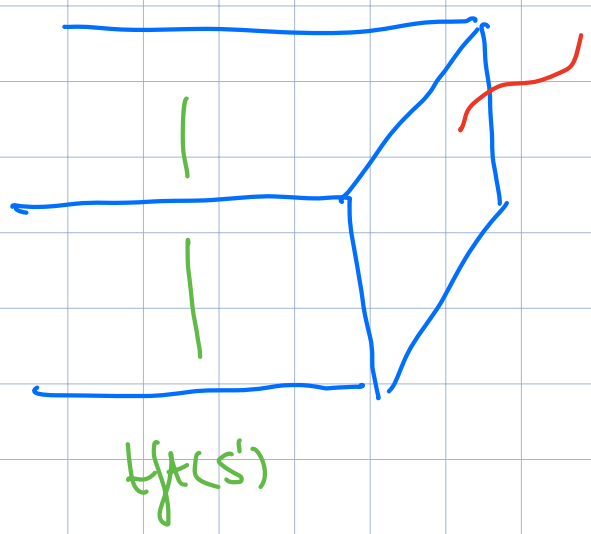


Willerton, Parmesan.

$$\tau(\omega)(g_1, g_2; \gamma) = \omega(\gamma, g_1, g_2)\,\omega(g_1, {}^{g_1}\gamma, g_2)^{-1}\,\omega(g_1, g_2, {}^{g_1g_2}\gamma)$$

2. Boundary conditions

2.1 General Turaev-Viro



bdry Wilson lines for a given boundary condition: spherical fusion cat. $\mathcal{W}_a$.

$$\begin{tikzcd} & Z(\mathcal{W}_a) \arrow[d, "\text{forget}"] \\ \mathrm{tft}(S^1) \arrow[ur, "\cong", "\text{bdr}"'] \arrow[r, "\otimes"'] & \mathcal{W}_a \end{tikzcd}$$

General Wilson lines: $\mathcal{W}_b \in \mathcal{W}_a\text{-Mod}$

$\mathrm{Fun}_{\mathcal{W}_a}(\mathcal{W}_{b_1}, \mathcal{W}_{b_2})$

bc = bicategory $\mathcal{W}_a$-Mod

Consistency check: $Z(\mathrm{Fun}_{\mathcal{W}_a}(\mathcal{W}_b, \mathcal{W}_b)) \cong Z(\mathcal{W}_a) \cong \mathrm{tft}(S^1)$
(pivotal Morita theory)

Surface defects: $\mathcal{C}_1 \boxtimes \mathcal{C}_2^{rev} \cong Z(\mathcal{W}_a)$

2.2 Module categories over vect_G^ω [Ostrik]

Associator

$$\alpha_{g_1 g_2 g_3} : (V_{g_1} \otimes V_{g_2}) \otimes V_{g_3} \longrightarrow V_{g_1} \otimes (V_{g_2} \otimes V_{g_3})$$

$$u \otimes v \otimes w \mapsto \omega(g_1, g_2, g_3)\, u \otimes v \otimes w$$

Subgroup $H \leq G$ + $\Theta \in C^2(H, \mathbb{C}^\times)$ s.t. $d\Theta = \omega$

$A_{H,\Theta} := \mathbb{C}_\Theta[H]$ special symm. Frobenius algebra in vect_G^ω

$\rightsquigarrow$ Module cat $A_{H,\Theta}\text{-mod} = \mathcal{M}(H,\Theta)$ over vect_G^ω

How get this geometrically from bundles?

2.3 Relative Bundles

<u>Def</u>: $H \xrightarrow{i} G$ morph of finite groups

$$\mathrm{Ind}_i : \mathrm{Bun}_H(X) \to \mathrm{Bun}_G(X)$$

$$P_H \longmapsto P_H \times_G G$$

- i inj $\Rightarrow$ Ind_i inj. on morphisms
- Commutes with pullback $\quad f : X_1 \to X_2$

$$\mathrm{Ind}_i\, f^* P_H^{(2)} = f^* \,\mathrm{Ind}_i\, P_H^{(2)}$$

morphism of stacks $\quad \mathrm{Ind}_i : \mathrm{Bun}_H \to \mathrm{Bun}_G$

<u>Def</u> : Relative mfd $\quad Y \xrightarrow{j} X$. (E.g. boundaries, defects)

Morphisms

$$\begin{tikzcd} Y_1 \arrow[r, "f_Y"] \arrow[d, "j_1"'] & Y_2 \arrow[d, "j_2"] \\ X_1 \arrow[r, "f_X"'] & X_2 \end{tikzcd}$$

<u>Def</u> Relative bundle $\quad \mathrm{Bun}_{H \xrightarrow{i} G}(Y \xrightarrow{j} X)$

$$= \left\{ \begin{array}{ccc} P_G & & P_H \\ \downarrow & & \downarrow \\ X & & Y \end{array} \qquad \alpha : \mathrm{Ind}_H^G P_H \xrightarrow{\sim} j^* P_G \right\}$$

<u>Example</u>

$\mathcal{T} =$

$i_1 : H_1 \to G_1$

$i_{12} : H_{12} \to G_1 \times G_2$

$i_2 : H_2 \to G_2$

$$\mathrm{Bun}(\mathcal{I}) = G_1 \times G_2 \backslash G_1 \times G_1 \times G_2 \times G_2 / H_1 \times H_{12} \times H_2$$

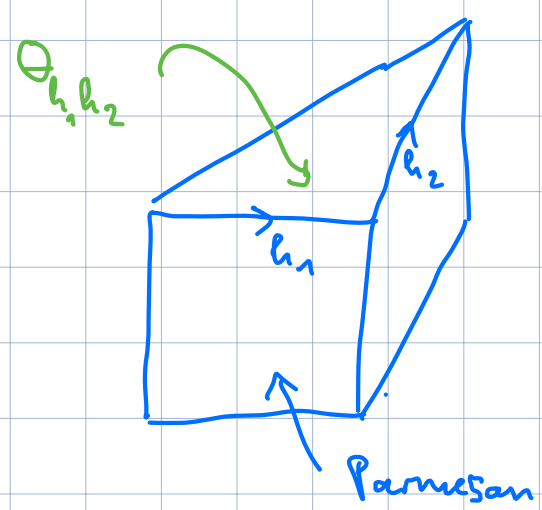
Cohomological data

$\omega_a \in Z^3(G_a, \mathbb{C}^*)$ bulk Lagrangian

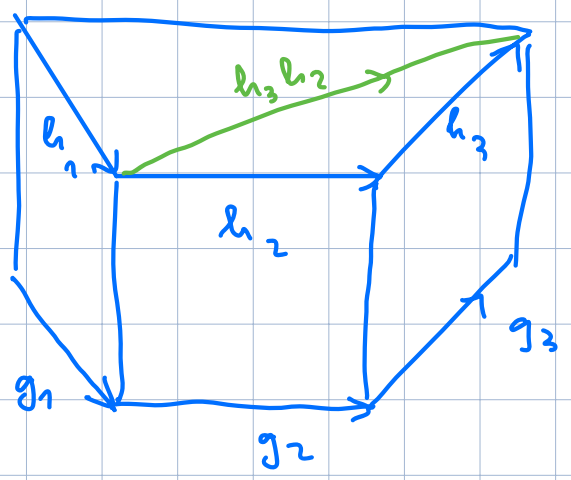
$\Theta_a \in C^2(H_a, \mathbb{C}^*)$ $\qquad d\Theta_a = i_a^* \omega_a$

$\Theta_{12} \in C^2(H_{12}, \mathbb{C}^*)$ $\qquad d\Theta_{12} = i_{12}^*(\omega_1 \omega_2^{-1})$

Transgress to 2-cocycle on groupoid $\mathrm{Bun}(\mathcal{I})$

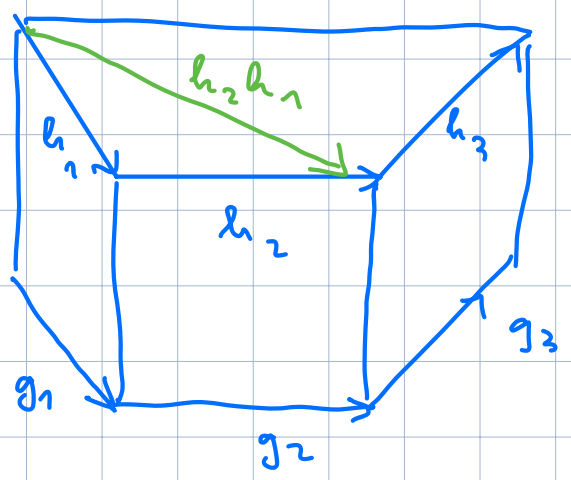


Cocycle condition:



2 triangles + 6 tetrahedra

$\|$



2 triangles + 6 tetrahedra

Equality $\iff$ $d\theta = i^* \omega$ [p. 27]

= cocycle condition

$$\begin{array}{ccc} & & \omega \\ & & \downarrow \\ P_G & M \longrightarrow & BG \\ & & \downarrow \eta \\ P_H & : \Sigma \longrightarrow & BH \end{array}$$

$i^* \omega = d\theta$

Structure familiar from σ-models:

η = "twisted gerbe"

Important check:

$$\mathrm{tr}\left(\mathrm{I}_G^{H_2}{}_{H_1} \right) \cong \mathrm{Fun}_{\mathrm{vect}_G^\omega}\left(\mathcal{M}(H_1, \theta_1), \mathcal{M}(H_2, \theta_2) \right)$$

2.4 Symmetries for abelian DW

Special case $G = A$ abelian, $\omega = 1$

$$\mathrm{Br}\,\mathrm{Pic}(\mathrm{vect}_A) = O_q(A \oplus A^*) \quad \text{w/ } q(g,\chi) = \chi(g)$$

non-degenerate.

Obvious symmetries:

1) of Bun_A $\quad \varphi \in \mathrm{Aut}(\mathrm{Bun}_A) = \mathrm{Aut}(A)$

Subgroup $\quad H_\varphi = \mathrm{graph}\,\varphi \subset A \oplus A \qquad \Theta = 1$

Braided eqn $\quad \varphi \oplus (\varphi^*)^{-1} : A \oplus A^* \longrightarrow A \oplus A^*$

2) Automorphisms of CS 2-gerbe = 1-gerbe on Bun_A

= "B-field"

$$H^2(A, \mathbb{C}^*) \xrightarrow{\sim} AB(A, \mathbb{C}^*) \ni \beta$$

Subgroup $\quad A_{diag} \subset A \oplus A \qquad \Theta = \beta$

Braided equivalence: $A \oplus A^* \longrightarrow A \oplus A^*$

$$(g, \chi) \longmapsto (g, \chi \cdot \beta(g, -))$$

3) Partial e-m dualities

<u>Ex</u> A cyclic, fix $\delta : A \to A^*$

Braided equivalence: $A \oplus A^* \longrightarrow A \oplus A^*$

$$(g, \chi) \longmapsto (\delta^{-1}\chi, \delta g)$$

Subgroup $A_{diag} \subset A \oplus A$

$$\beta(a_1, a_2) = \frac{\delta(a_1)(a_2)}{\delta(a_2)(a_1)} \in AB(A, \mathbb{C}^*)$$

<u>Thm</u> [FPSV '14] Set of generators for $\mathrm{BrPic}(\mathrm{vect}_A)$.

3. Outlook

- DW as a laboratory
- Other dimensions? Higher codimension defect?